



DATA ANALYSIS & BUSINESS INTELLIGENCE PROJECT

BMW Sales Dashboard

From Raw Data to Business Insight — Pandas Cleaning & Power BI Dashboard

PROJECT TEAM

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A complete walkthrough of the project, from raw data to business recommendations.

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► **Tools Used**

- Python 3
- Pandas
- Power BI Desktop
- DAX
- Star Schema Modeling

Project Overview

Turning a raw sales export into a decision-ready dashboard.

This project analyzes five years (2019–2023) of global BMW sales data to uncover revenue trends, identify top and bottom performing models, compare sales channels, and support data-driven business decisions. The work was split into two connected stages, matching a real-world analytics workflow:

Pandas

CLEANING & TRANSFORMATION

Power BI

MODELING & DASHBOARD

DAX

BUSINESS MEASURES

► Stage 1 — Data Cleaning (Python / Pandas)

The raw CSV export was inspected for structural issues, missing values, incorrect data types, and duplicates. A reusable cleaning script ([clean_bmw_sales.py](#)) was written to turn the messy export into an analysis-ready table.

► Stage 2 — Dashboard (Power BI)

The cleaned data was loaded into Power BI Desktop and modeled as a **star schema** (one fact table + four dimension tables), then visualized across an executive Dashboard page and a detailed "All Models" drill-down page, using DAX measures for growth, variance, and time intelligence.

► Note on Scope

Reference tutorials for this dataset (e.g. the "From Data to Dashboard" Power BI walkthrough) typically clean the data *inside* Power Query. In this project, all cleaning and transformation was deliberately done in **Pandas before** the data ever reached Power BI — Power BI was used purely for modeling, DAX measures, and visualization.

The Dataset

What the raw data looked like before any cleaning.

The source file, **BMW_Sales_Data.csv**, contains transaction-level BMW sales records collected across five regions and 23 countries between **2019 and 2023**. Each row represents one sale of a BMW model on a given date.

5,778
RAW ROWS

8
COLUMNS

26
BMW MODELS

23
COUNTRIES

5
REGIONS

► Raw Column Reference

COLUMN	ORIGINAL TYPE	DESCRIPTION
Date	Text (DD/MM/YYYY)	Date the sale took place
Year	Float	Sale year (redundant with Date, and unreliable — see cleaning)
Model	Text	BMW model sold (e.g. BMW X5, BMW M4, BMW i8)
Revenue	Float	Total revenue generated by that sale
Quantity Sold	Float	Number of units sold in that transaction (1-5)
Region	Text	Africa, Asia, Europe, North America, South America
Country	Text	Country of sale (23 unique countries)
Channel	Text	Wholesale, Dealership, or Online

► Categorical Coverage

Models (26): BMW 1-8 Series, X1-X7, Z4, M2/M3/M4/M5/M8, and the electric i3/i4/i8/iX/iX3 line-up.

Channels (3): Wholesale Dealership Online

Regions (5): Africa Asia Europe North America South America

Data Cleaning & Transformation

Every problem we found in the raw file, and how Pandas fixed it.

► Problems Found in the Raw Data

- 1 **778 completely empty rows** (13.5% of the file) — every single column blank. These were almost certainly export artifacts, not real transactions.
- 2 **Dates stored as plain text** in **DD/MM/YYYY** format instead of a real date type — this silently blocks any time-based analysis (trends, YoY growth, time intelligence).
- 3 **Wrong data types** — **Year** and **Quantity Sold** were stored as decimals (e.g. 2019.0) instead of whole numbers, a side effect of the blank rows forcing the column into a float type.
- 4 **No price-per-unit metric** existed in the raw data, even though it's one of the most useful KPIs for comparing models.
- 5 **Untrimmed / inconsistent text** in the Model, Region, Country and Channel columns needed to be standardized before they could safely be used as dashboard filters.

► What We Checked but Did Not Find

We also tested for negative or zero revenue/quantity values, duplicate real transactions, and spelling inconsistencies in categorical fields — **none were found**. Once the blank rows were removed, the underlying data was consistent and required no further correction.

► Cleaning Steps Applied (Pandas)

1. Dropped the 778 fully-empty rows, and any remaining rows missing a critical field.
2. Parsed **Date** with **pd.to_datetime()** using the **DD/MM/YYYY** format.
3. Re-derived **Year** directly from the cleaned Date column (safer than trusting the original) and cast it to integer.
4. Cast **Quantity Sold** to integer and rounded **Revenue** to 2 decimals.
5. Added a new column — **Revenue Per Unit** = Revenue ÷ Quantity Sold — for price-level analysis.
6. Stripped whitespace and standardized casing across all text columns.
7. Dropped exact duplicate rows and sorted the table chronologically.

```
# core of the cleaning script
df = df.dropna(how="all")
df = df.dropna(subset=critical_cols)
df["Date"] = pd.to_datetime(df["Date"], format="%d/%m/%Y")
df["Year"] = df["Date"].dt.year.astype(int)
df["Quantity Sold"] = df["Quantity Sold"].astype(int)
df["Revenue Per Unit"] = (df["Revenue"] / df["Quantity Sold"]).round(2)
```

5,778

ROWS BEFORE

-778

ROWS REMOVED (13.5%)

5,000

ROWS AFTER CLEANING

+1

COLUMN ADDED

Analysis Questions & KPIs

The business questions this project was designed to answer.

► Business Questions

- 1 How has total revenue trended year over year (2019–2023)? Is the brand growing, shrinking, or flat?
- 2 Which BMW models generate the most revenue, and are they the same models that sell the most units?
- 3 Which sales channel — Wholesale, Dealership, or Online — contributes the most (and least) to revenue?
- 4 Which regions and countries drive the largest share of sales, and are any regions underperforming?
- 5 Is there seasonality in sales — do certain months consistently outperform others?
- 6 What is the average price per unit, and how much does it vary between models?
- 7 Which models or markets are underperforming and may need a pricing, marketing, or inventory response?

► KPIs Tracked

KPI	DEFINITION	WHY IT MATTERS
Total Revenue	Sum of Revenue across all sales	Headline measure of business performance
Total Units Sold	Sum of Quantity Sold	Measures demand independent of price
Revenue Growth %	YoY / MoM % change in revenue	Shows momentum — growing or slowing
Average Price / Unit	Revenue Per Unit, averaged	Signals brand/model positioning
Revenue by Channel	Revenue split across Wholesale / Dealership / Online	Guides channel investment
Revenue by Region / Country	Geographic breakdown of revenue	Guides market prioritization
Top / Bottom Models	Ranked model performance	Directs product & marketing focus

These questions and KPIs directly shaped which visuals were built into the Power BI dashboard (Section 06) and are answered with real numbers from the cleaned dataset in Section 07.

Dashboard & Data Model

How the cleaned data was structured and measured inside Power BI.

The cleaned CSV was loaded into Power BI Desktop and modeled as a **star schema** — one central fact table surrounded by four dimension tables — which keeps the model fast, simple to filter, and ready for time-intelligence DAX functions.

► Fact Table — Fact_Table

COLUMN	ROLE
Date	Transaction date
ModelID / CountryID / ChannelID	Foreign keys to dimension tables
Quantity Sold	Units sold
Total / Unit Price	Revenue and price per unit

► Dimension Tables

TABLE	KEY FIELDS
dimModel	ModelID, Model, img (model image)
dimCountry	CountryID, Country, Region, Flag
dimChannel	ChannelID, Channel
Calendar	Date, Month, MonthNum, Quarter, Weekday

All relationships are one-to-many, flowing from each dimension table into **Fact_Table**.

► Key DAX Measures

Revenue

Revenue Growth %

Revenue Variance %

Average Price

Previous Month Revenue

Same Period Last Year

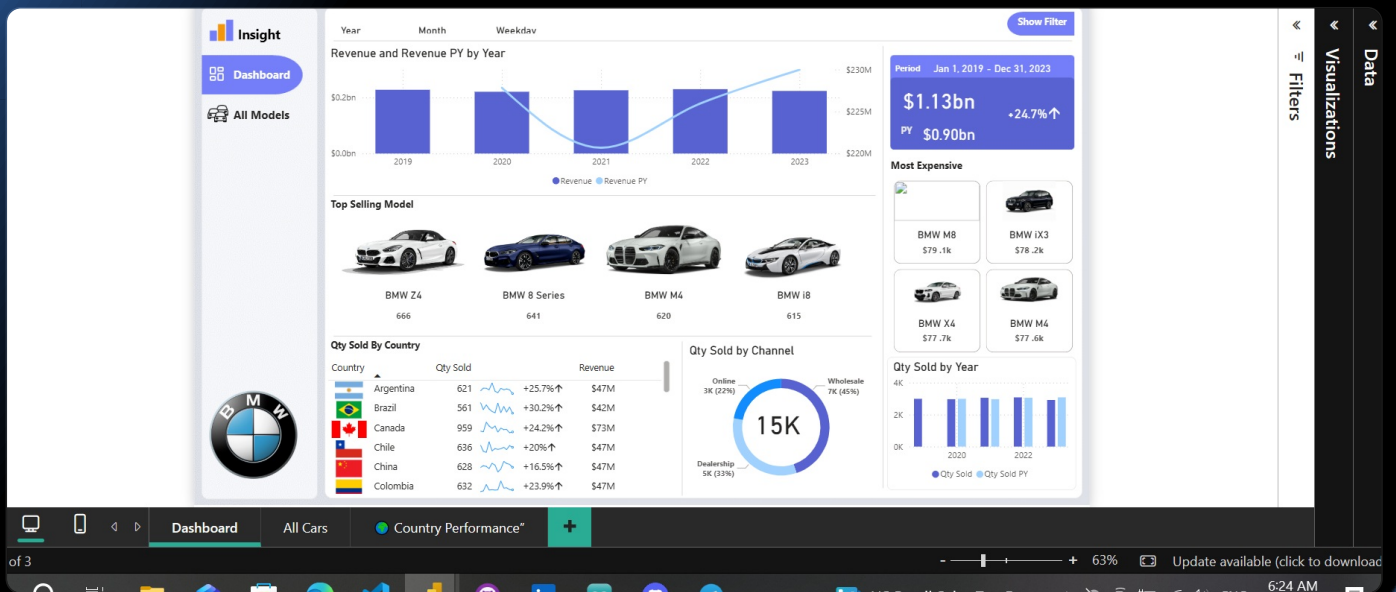
Built with Power BI time-intelligence functions including **CALCULATE**, **DATEADD**, **TOTALYTD**, and **PREVIOUSMONTH**, plus arrow-style variance indicators to flag under/over-performing periods at a glance.

► Navigation & UX

- Interactive Year / Month / Weekday filters
- Bookmarks and action buttons for page-to-page navigation
- Growth arrow indicators (↑ / ↓) for quick visual scanning
- Separate pages for executive overview, per-model drill-down, and country performance

Dashboard Screenshots

The final Power BI report — Executive Dashboard page.



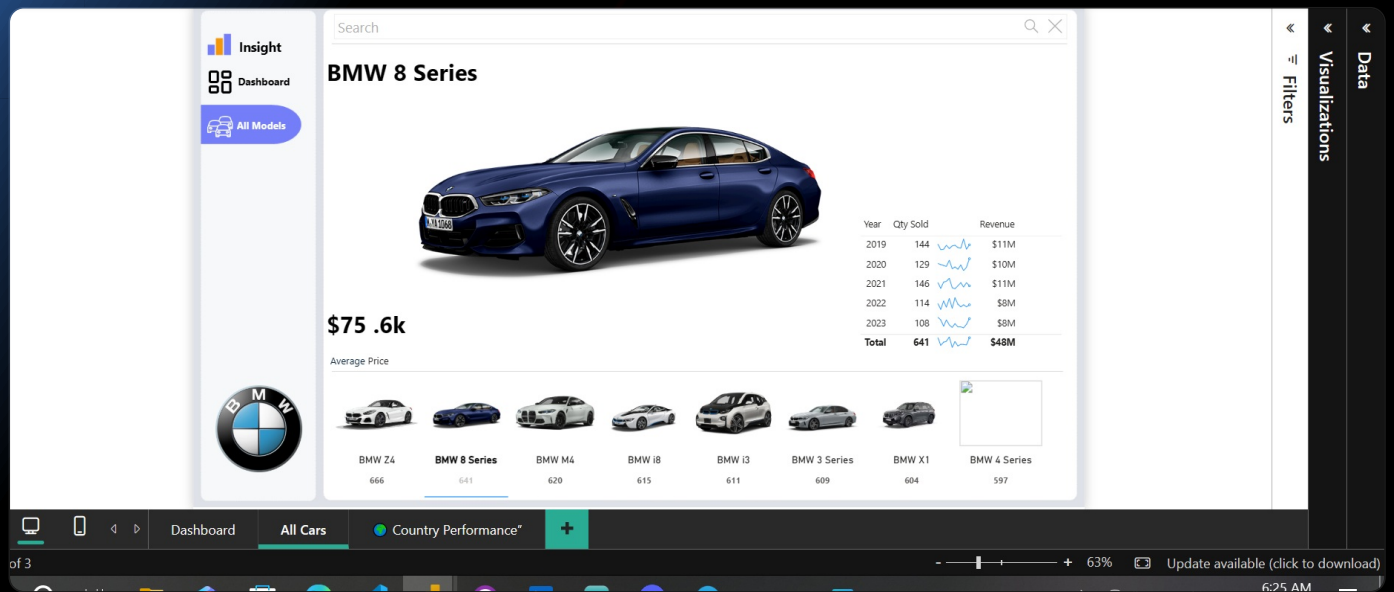
Executive Dashboard — global revenue, top models, and channel breakdown

► What this page shows

- **Revenue & Revenue PY by Year** — answers Question 1 (year-over-year trend), with a \$1.13bn cumulative total and +24.7% vs. the prior comparison period shown in the KPI card.
- **Top Selling Model** gallery — answers Question 2, ranking models by units sold (BMW Z4 leads at 666 units).
- **Most Expensive** card — surfaces the highest average-price models, supporting Question 6.
- **Qty Sold by Country** table with sparklines and Revenue — answers Question 4 at a country level.
- **Qty Sold by Channel** donut — answers Question 3 (Wholesale 45%, Dealership 33%, Online 22%).
- **Qty Sold by Year** bar chart — a secondary view supporting Question 1 in unit terms, not just revenue.

Dashboard Screenshots

The final Power BI report — All Models drill-down page.



All Models page — per-model detail, shown here for the BMW 8 Series

What this page shows

- A searchable model gallery lets a user click any BMW model and drill into its own performance.
- A **Year × Qty Sold × Revenue** table with per-year sparklines directly answers Question 1 and Question 7 at the individual model level — e.g. the BMW 8 Series peaked in 2021 (146 units) and has declined every year since, to 108 units in 2023, flagging it as a model to watch.
- The **Average Price** card supports Question 6, and the model strip along the bottom lets stakeholders compare relative popularity across all 26 models in one glance (answers Question 2).

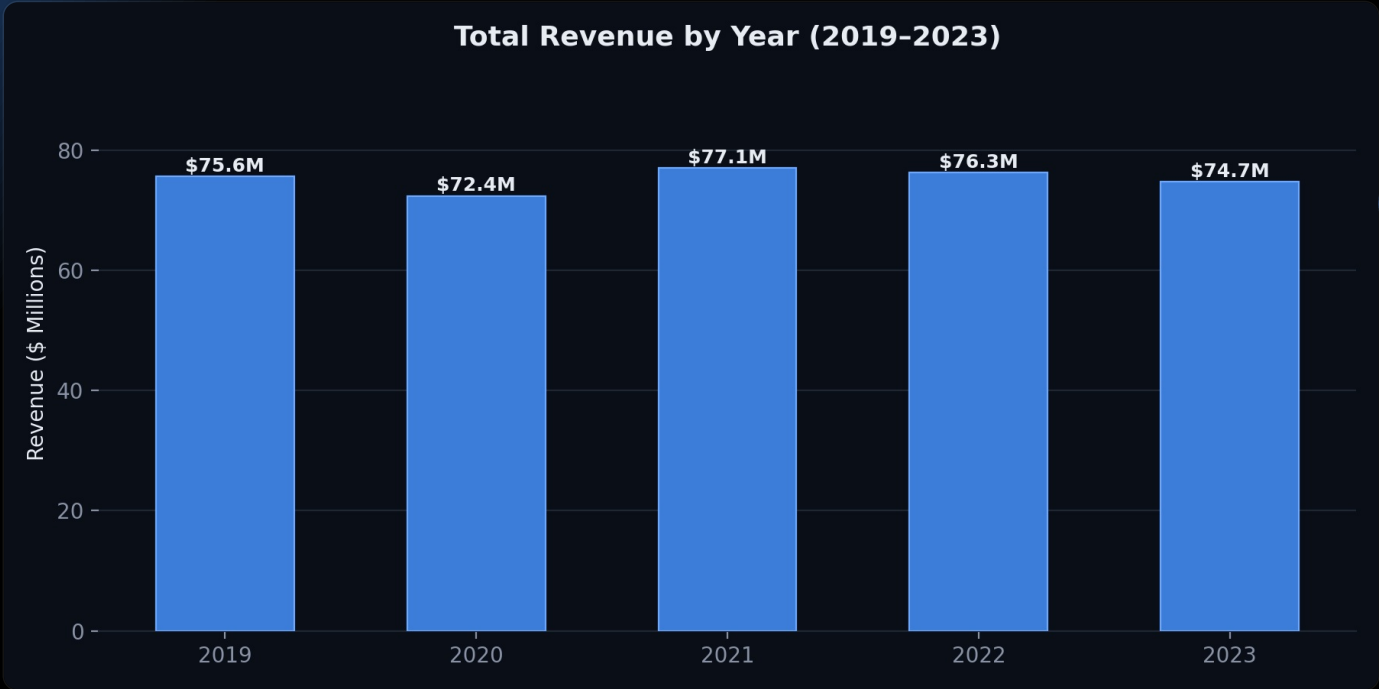
Together, these two pages plus a third *Country Performance* page (not pictured) give stakeholders both a bird's-eye executive view and a click-through path to model- and country-level detail — covering all seven analysis questions from Section 04.

Key Findings

Answering our analysis questions with the cleaned data.

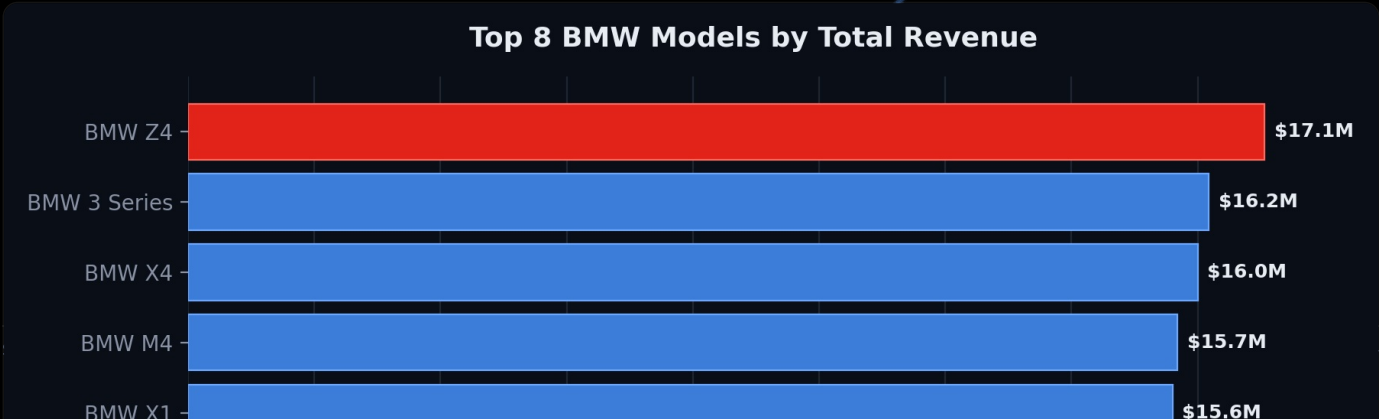
\$376.1M TOTAL REVENUE	15,002 UNITS SOLD	\$34,187 AVG. PRICE / UNIT	-2.0% 2023 YOY REVENUE
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► Q1 — Is revenue growing year over year?



Revenue is essentially **flat to slightly declining** across the five-year window: a dip in 2020 (−4.3%), a rebound in 2021 (+6.4%), then two consecutive years of mild decline (−1.0% in 2022, −2.0% in 2023). The brand isn't shrinking dramatically, but it has plateaued since its 2021 peak.

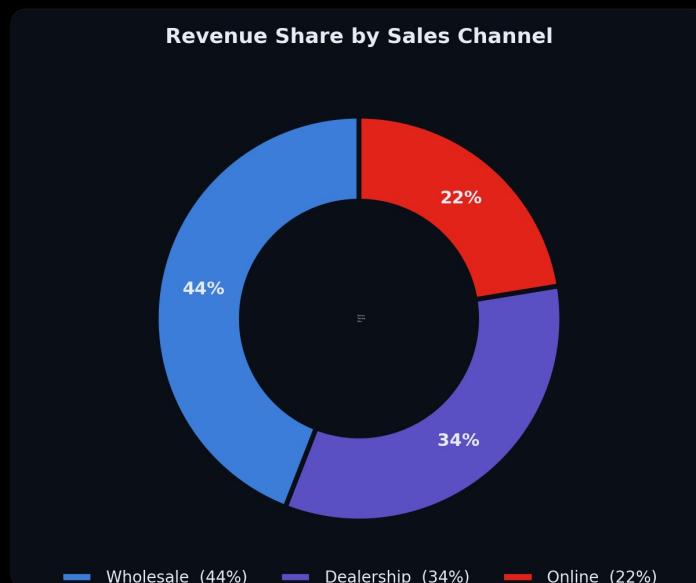
► Q2 — Which models lead, and do revenue and volume agree?



Key Findings (continued)

Channel and regional performance.

► Q3 — Which channel drives the most revenue?



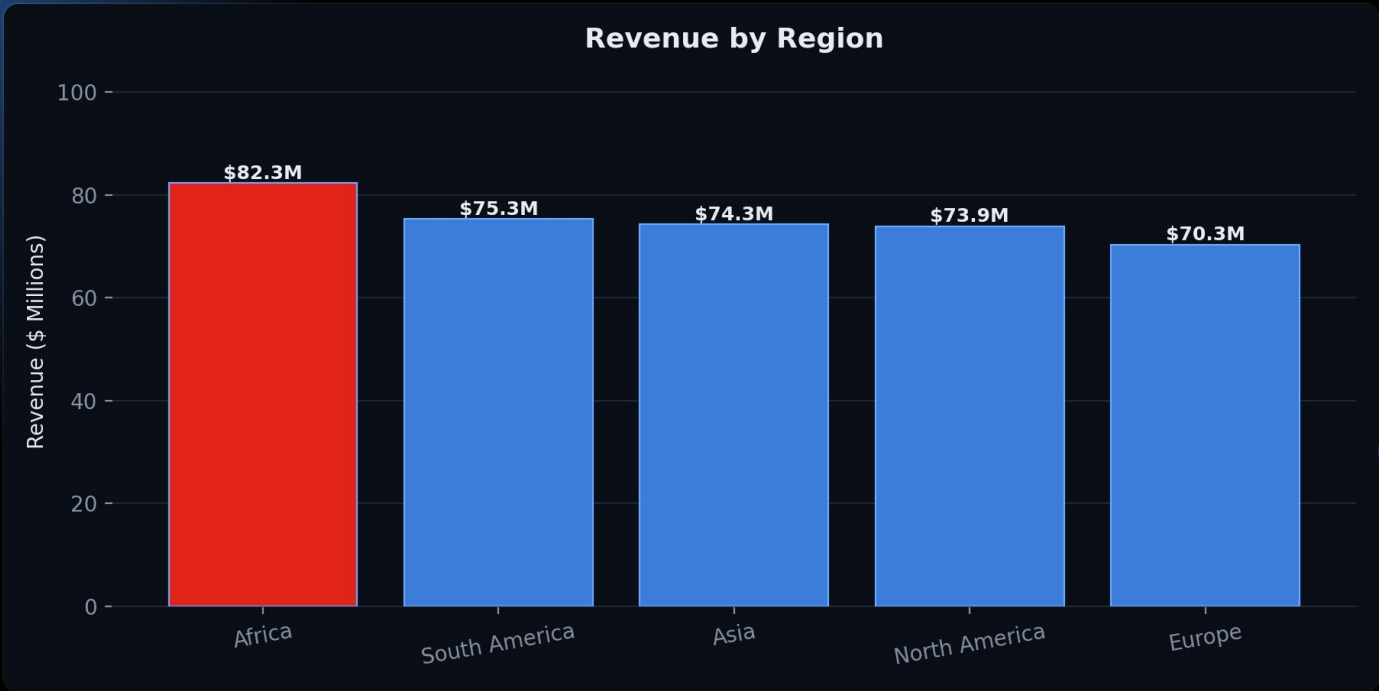
Wholesale dominates at **44.1%** of revenue (\$165.8M), followed by Dealership at 33.5% (\$126.0M). **Online is the smallest channel at just 22.4%** (\$84.3M) — a clear gap relative to how e-commerce typically performs in other industries.

A heavy tilt toward Wholesale means BMW's revenue is concentrated in bulk, relationship-driven deals rather than direct-to-consumer sales — a structural insight that directly informs Recommendation 01 in Section 08.

Key Findings (continued)

Geography, seasonality, and pricing.

► Q4 — Which regions/countries lead?



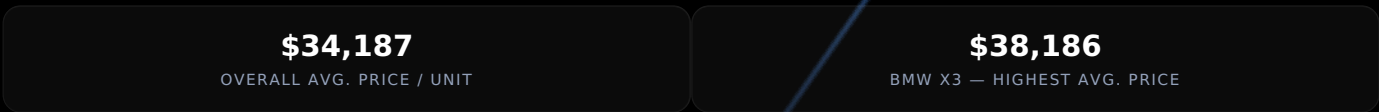
Revenue is fairly balanced geographically: **Africa leads** with 21.9% of revenue (\$82.3M), while **Europe trails** all regions at 18.7% (\$70.3M) despite being BMW's home market. At the country level, **Mexico** (\$25.18M) and the **United States** (\$25.10M) are the top two individual markets.

► Q5 — Is there seasonality?



Revenue peaks in **August** and dips lowest in **February** — a modest, manageable seasonal swing rather than a dramatic cycle, but still useful for planning inventory and promotions (see Recommendation 04).

► Q6 — How much does price vary by model?



The **BMW X3** commands the highest average price in the range (~\$38,186), followed closely by the M8 and X4 — the SUV and M-performance lines clearly carry the strongest pricing power across the BMW line-up.

Recommendations

What we'd advise the business to do based on this analysis.

01 Grow the Online channel

Online is the smallest channel at 22.4% of revenue despite requiring the least overhead. Investing in a stronger online configurator/checkout experience could shift volume away from Wholesale dependence and improve margins.

02 Double down on the Z4 — and investigate the 5 Series

The Z4 wins on both revenue and volume; give it priority in marketing spend and inventory allocation. In contrast, the 5 Series is the weakest revenue performer — worth a pricing, trim, or positioning review to understand why.

03 Re-energize the European market

Europe generates the lowest regional revenue share despite being BMW's home turf — a signal worth investigating for competitive pressure, and a candidate for a refreshed regional campaign.

04 Plan inventory and promotions around the August peak

With August the strongest month and February the weakest, align stock builds and launch timing with August/September demand, and use targeted promotions to lift the February trough.

05 Push the EV line-up to reverse the post-2021 plateau

Revenue has softened for two straight years after the 2021 rebound. A renewed push behind the i3/i4/i8/iX/iX3 electric range is a natural growth lever to break out of the current plateau.

06 Use premium SUVs to lift average revenue per unit

The X3, M8, and X4 command the highest prices in the range. Bundling higher trims or upselling these models is the most direct way to raise average revenue per unit fleet-wide.

Conclusion

Wrapping up the project.

This project walked a BMW sales dataset through the full analytics pipeline — from a raw, imperfect CSV export to a polished, interactive Power BI dashboard. The **Pandas cleaning stage** revealed that the dataset's problems were structural rather than "dirty": 778 blank rows, text-formatted dates, and float-typed integer columns — all fixed with a short, reusable cleaning script, with no duplicate transactions or invalid values found underneath.

On the **Power BI side**, modeling the cleaned data as a star schema with dedicated Model, Country, Channel, and Calendar dimension tables made it possible to build fast, filterable visuals and proper time-intelligence DAX measures — turning seven business questions into an executive dashboard that any stakeholder can explore in seconds.

The **analysis itself** tells a coherent story: BMW's global sales are large (\$376M across five years) but essentially **flat since 2021**, revenue is **over-reliant on Wholesale** relative to Online, and performance varies meaningfully by model (Z4 excelling, 5 Series lagging) and by region (Africa strong, Europe surprisingly weak). Each of these findings maps directly to a concrete recommendation in Section 08.

► In short

Clean data made reliable analysis possible, and a well-modeled dashboard turned that analysis into insights a business can actually act on — which was the goal of this project from the start.

► Deliverables

- **clean_bmw_sales.py** — the Pandas cleaning script
- **BMW_Sales_Data_Clean.csv** — the cleaned dataset (5,000 rows)
- **BMW_Dashboard.pbix** — the Power BI dashboard file
- This report — full project documentation